

Exercises

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Surname, First name

5ARC0 Human and ethical aspects of AI
Final exam Q1

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☒ a	☐ b	☐ c	☒ d	☒ e	☐ f	→ a, d
☒ a	☐ b	☐ c	☐ d	☒ e	☐ f	→ e
☒ a	☐ b	☒ c	☒ d	☐ e	☐ f	→ c

Answer multiple-choice questions as shown in the example. Circular checkboxes have only one correct answer. Square checkboxes may have multiple correct answers.

Particular Ans on paper exam instructions

- Write in a black or blue pen.
- You answer open-ended questions by using the text box. Provide your answers on the papers inside the answer box underneath a question. ***If you need more space for your answers, use the extra space at the end of the exam, and clearly indicate there which question you continue answering. In the text box of the particular question, clearly state that you proceed with your answer on a different page.***
- Hand in all pages. Do not remove the staple. If you remove it anyhow, check that you hand in all pages.

Dear student,

You're about to take an exam. Write down your name and your student ID at the appropriate places above. Make sure that you enter your student ID by fully coloring the appropriate boxes. On the examination attendance card, you fill in the PDF number. You can find the correct number on the top of the first page of your exam (e.g. 1234.pdf).

Please read the following information carefully:

Date exam: 4 November 2025

Start time 09.00

End time: 12.00 (+30 minutes for time extension students)



Number of questions: 27

Maximum number of points/distribution of points over questions: 50 points total.

Method of determining the final grade: actual score/ 5

Answering style: short essay (single paragraph), short answer, fill in the blank, matching, multiple answer, multiple choice

Permitted examination aids

- Scrap paper (fully blank)
- Dictionaries. If yes, please specify: English-other language and/ or vice versa

Important:

- You are only permitted to visit the toilets under supervision
- Examination scripts (fully completed examination paper, stating name, student number, etc.) must always be handed in
- The house rules must be observed during the examination
- The instructions of subject experts and invigilators must be followed
- Keep your work place as clean as possible: put pencil case and breadbox away, limit snacks and drinks
- You are not permitted to share examination aids or lend them to each other
- Do not communicate with any other person by any means

During written examinations, the following actions will in any case be deemed to constitute fraud or attempted fraud:

- using another person's proof of identity/campus card (student identity card)
- having a mobile telephone or any other type of media-carrying device on your desk or in your clothes
- using, or attempting to use, unauthorized resources and aids, such as the internet, a mobile telephone, smartwatch, smart glasses etc.
- having any paper at hand other than that provided by TU/e, unless stated otherwise
- copying (in any form)
- visiting the toilet (or going outside) without permission or supervision

First-year bachelor students: The final grade for this exam will be announced no later than fifteen working days after the date of this exam, unless this exam takes place in Q4 or the interim period. For Q4 final exams, grades will be announced within five working days after the end of the Q4 final test period. For interim period final exams, grades will be announced no later than five working days before September 1.

All other students: Generally, the final grade for this exam will be announced no later than fifteen working days after the date of this examination. Specifically for bachelor exams administered in the interim period, exam grades will be announced no later than five working days before September 1.

You can start the exam now, good luck!

Main Part

- 2p **1a** Which of the following kinds of AI research would typically be subject to institutionally or legally mandatory ethics review? (Choose all that apply.)
- ☐ Research on automating industrial fault detection using AI
 - ☐ Market analysis using AI at a Dutch tech company.
 - ☐ Research at TU/e that concerns big data analysis of student learning to give individual student feedback
 - ☐ Research at a hospital on a new kind of AI-powered medical device
- 1p **1b** What value is referred to by the “Plus One” in the “Four Principles Plus One” ethics framework used by the EU to evaluate artificial intelligence? (Choose one.)
- ☐ a Sustainability
 - ☐ b Accountability
 - ☐ c Explicability
 - ☐ d Responsibility
- 2p **1c** Consider the **right to employment** articulated in the Universal Declaration of Human Rights: “the right to work, to free choice of employment, to just and favourable conditions of work and to protection against unemployment” (Article 23, Part 1). Which of the following are true statements about this right?
- ☐ The duty-holders include the government.
 - ☐ The right-holder includes any adult human that is able to work.
 - ☐ Widespread unemployment does not threaten this right.
 - ☐ This is a purely negative right.

2p **1d** Which of the following is a defining characteristic of machine learning, as discussed in Chapter 1 of the textbook? (Choose all that apply.)

- ☐ It gets better at a task without having its programming explicitly changed by a human.
- ☐ Its performance at a specific task improves with experience.
- ☐ It passes the Turing Test, allowing it to simulate human interaction.
- ☐ It only uses neural networks.

1p **1e** What two main kinds of reasoning are carried out in artificial intelligence? (Choose one.)

- ☐ a Valid and sound reasoning
- ☐ b Generalization and specification
- ☐ c Ethical and normative
- ☐ d Probabilistic and deductive

2p **1f** Suppose we are trying to formulate an argument about an AI model:

1. _____
2. Unless the model is explainable, there will be a high risk that the model cannot be adjusted to fit new data.
3. Therefore, the model should be made explainable.

What is the missing premise (1) that would make this a valid, non-circular argument?



1p **1g** What is the main claim supported by the Chinese Room thought experiment?

- ☐ a Embodied cognition is necessary for intelligence
- ☐ b Syntax alone is sufficient for semantics
- ☐ c AI systems can process natural language as well as humans
- ☐ d Intelligence is more than symbol manipulation and computation

1p **1h** Which of the following best captures the main idea behind "Design for Values"?

- ☐ a Narrow technical implementation of values such as fairness in AI models
- ☐ b Avoiding bias during the selection of ethical values
- ☐ c Embedding ethical principles after deployment to ensure compliance
- ☐ d Translating ethical values into design requirements

2p **1i** Which of the following illustrates a tension between values in the design of AI-based systems?

- ☐ Accuracy vs. fairness when optimizing predictive performance
- ☐ Privacy vs. fairness when datasets require demographic attributes
- ☐ Transparency vs. sustainability when publishing open-source models

4p **1j** In Chapter 4 of the textbook, Naudts and Vedder introduce two conceptions of justice. The first is a **relational** view of justice that defines it as rooted in social relationships, processes, and structures rather than merely in outcomes or procedures. The second is a **distributive** view of justice that aims at equality or proportionality in distribution of outcomes. Which of these two views can be more easily implemented as algorithmic fairness, and why? As part of your answer, define algorithmic fairness, and give a short motivation for your answer.

- 1p **1k** What is the central concern underlying the claim that there is a “responsibility gap” in AI systems?
- ☐ (a) AI systems cannot be held morally responsible because they lack consciousness
 - ☐ (b) Responsibility gaps occur when AI systems act unlawfully
 - ☐ (c) AI systems are inherently opaque and not explainable
 - ☐ (d) Autonomous AI systems create situations where one cannot assign responsibility to any human agent
- 1p **1l** The authors of Chapter 5 of the textbook argue that responsibility gaps are not morally problematic. Which of the following statements best corresponds to their argumentation for this thesis?
- ☐ (a) We can assign responsibility to AI systems themselves.
 - ☐ (b) We can eliminate responsibility gaps by minimizing the use of fully autonomous AI.
 - ☐ (c) Forward looking responsibility, establishing accountability and harm prevention, can be achieved in the domain of AI.
 - ☐ (d) The law and moral codes do not recognize the responsibility of AI based systems.
- 2p **1m** Which of the following are necessary conditions for assigning backward-looking moral responsibility to an individual in cases where there has been a harmful outcome? (Choose all that apply.)
- ☐ The individual could have foreseen the possibility of harm
 - ☐ The individual acted freely
 - ☐ The individual was part of a collective
 - ☐ The risk was acceptable



- 2p **1n** In Chapter 6 of the textbook, Hasselbalch and Van Wynsberghe argue that data leakage resembles environmental pollution because both create a certain kind of harm. What main characteristic does this kind of harm have, and why does it matter for the strategy we should take to achieve more sustainable AI?

- 1p **1o** A global tech company is developing an AI-powered hiring tool that, in early testing, disproportionately excludes candidates from marginalized communities. The team of AI engineers responds by adding bias detection features.
At which level of AI sustainability governance does this intervention primarily occur?

- ☐ a The global level
- ☐ b The middle level
- ☐ c The individual level
- ☐ d The micro level

- 2p **1p** What is the basic principle of the Brundtland definition of sustainability, which underlies the anthropocentric view of the value of sustainability?



- 3p **1q** A bank uses a machine learning system to assess loan applications. For a teller entering customer data, vs. for a regulator auditing fairness, the system is opaque in different ways. On the basis of this example, what lesson does Zednik (author of "Solving the Black Box Problem") draw about the nature of opacity, and what does this imply for designing explainable AI? Keep your answer brief (1-2 sentences).

- 3p **1r** In your assignment project from this course, would there be a way of applying explainable AI so that it would help solve the black box problem for the client or user? Briefly explain how that would work; or, why it is not relevant. Include a few details from your assignment project.



The next two questions relate to the following text. In an article on the website of a military academy, Lena Trabucco writes the following: "MHC [Meaningful Human Control] is a loaded political concept that emerged from the debate on autonomous weapons. It generally refers to preserving human judgment and input while employing autonomous systems. Some advocates maintain that MHC is necessary for compliance with the law of armed conflict (LOAC). This position assumes that an AWS [Autonomous Weapons System] cannot comply with existing legal principles (notably distinction [i.e., **discriminacy**]).... However, [I argue to the contrary:] LOAC does *not* explicitly require human control..." (italics added).

- 3p **1s** Reconstruct the position that Trabucco is attacking: what argumentation is Trabucco attributing to "advocates" of MHC in this passage? Please include a normative and a descriptive premise in your reconstruction.

- 3p **1t** How does the version of MHC put forward by Santoni de Sio & van den Hoven, from the lecture and readings, differ from the account of MHC given by Trabucco?



3p **1u** Please match the three approaches/ requirements for building ethics into robotics from Goodall's article, to the methods on the right. There can be more than one match per approach/ requirement.

A. Rational Approach

1. Uses neural networks trained on crash scenarios and human feedback

B. Artificial Intelligence Approach

2. Applies principles like "injuries preferable to fatalities" within AI models

C. Natural Language Requirement

3. Translates complex neural network logic into symbolic rules expressed in human language

4. Applies Asimov's laws

5. Creates explanations of a complex deep learning model to give feedback to humans.



- 1p **1v** Please fill in the blank in the following statement about the approach adopted by Awad and colleagues in the Moral Machines project, to incorporate stakeholder perspectives into self-driving cars. The _____-in-the-loop framework emphasizes an iterative process where citizens, engineers, and regulators collaborate to align machine behavior with human values.

- 2p **1w** What approach to the empirical collection of stakeholder input is used within MIT's "Moral Machines" project? Describe the project in 2-3 sentences.

- 2p **1x** Which of the following propositions is part of the standard argument, found in Chalmers and Bostrom, that there is an existential risk from superintelligence?

- ☐ AI does not need to reach a level of intellect far beyond human cognitive performance, in order to pose existential risks.
- ☐ Intelligent agents would tend to adopt similar intermediate goals—such as self-preservation, resource acquisition, and power—because these help achieve almost any final goal.
- ☐ Humans will create alignment in superintelligent AI systems.
- ☐ Intelligence and overall goals are independent; any level of intelligence can be paired with almost any goal.



- 1p **1y** Which statement best captures the difference between **catastrophic risk** and **existential risk** in the context of AI?
- (a) Catastrophic risk refers to any harm on a large scale, while existential risk refers specifically to risks that could cause human extinction or permanently curtail humanity's potential.
 - (b) Catastrophic risk is about harms that can be mitigated, while existential risk is about long-term harms that cannot be mitigated.
 - (c) Catastrophic risk applies primarily to natural catastrophes, while existential risk applies primarily to technological threats.
 - (d) Catastrophic risk and existential risk are synonymous; both mean the same thing.
- 1p **1z** Which of the following is a common criticism of **longtermism** in the context of AI risk?
- (a) Superintelligence will lead only to good outcomes in the long term future.
 - (b) Future generations interests are not ethically important.
 - (c) It avoids the issue of value alignment in AGI systems.
 - (d) Focusing on longtermism leads to neglect of problems such as bias, discrimination, and opacity.
- 1p **1aa** In discussions of AI risk, what does **alignment** primarily refer to?
- (a) Ensuring that AI systems are computationally efficient to save resources.
 - (b) Ensuring that AI systems' goals and behaviors are very similar to human goals and values.
 - (c) Ensuring that different countries in the EU can engage in commercial and scientific exchange about AI.
 - (d) Ensuring that AI systems do not surpass humans in intelligence.

Ext

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